PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
KAGERA REGION
FORM FOUR MOCK EXAMINATIONS -2023
032/1 CHEMISTRY 1- MARKING SCHEME
YEAR 2023

SECTION A (15 Marks)

1.

i	ii	iii	Iv	v	Vi	Vii	viii	ix	x
A	D	B	E	B	C	A	D	D	A

1@10=10 Marks

2. Matching items **(06 marks @01mark)**

i. B
ii. C
iii. A
iv. D
v. G
vi. E

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3. a) i. Problem identification
ii. Asking questions
iii. Hypothesis formulation
iv. Experimentation **(05 mark @ I mark)**
v. Drawing conclusion

b) (i) Physical change
(ii) Chemical change
(iii) Physical change
(iv) Chemical change **(4 marks @ 1 mark)**

4. ⇛Faraday's first law of Electrolysis state that "The mass of a substance deposited or Liberated during electrolysis is directly proportional to the quantity of electricitypassed though the electrolyte" **(1.5 Mark)**

⇛Faraday's second law of electrolysis state that "When the same quantity of electricity is passed though solutions of different electrolytes, the mass of the substance liberated or deposited at the electrodes are directly proportional to the chemicalequivalents of the substances"**(1.5 Mark)**

(b) Soln.
Data given.
Volume of chlorine gas produced during electrolysis = $5dm^3$
Discharging equation of Cl.
$2Cl^- \rightarrow Cl_2 + 2e$
2F of Cl_2 = $22.4dm^3$ at STP
? = $5\ dm^3$

$= \frac{2\ F \times 5 dm^3}{22.4\ dm^3}$

$= 0.4464F$ **(02 Mark)**

Required:

(i) Volume of O_2

Discharge equation of Oxygen

$40H^-_{(ag)} \rightarrow 2H_2O + O_{2(g)} + 4\ e^-$ (Liberated at Anode)

$4F = 22.4\ dm^3$ of O_2 At STP

$0.4464F = ?$

$= \frac{0.4464\ F \times 22.4 dm^3}{4\ F}$

$= 2.49984 dm^3$

$= 2.5 dm^3$ **(02 Mark)**

<u>The volume of O_2 produced $= 2.5\ dm^3$</u>

(ii) Mass of Cu

Discharge equation of Cu

$Cu^{2+} + 2e \longrightarrow Cu$

$2F = 64g$ of Cu

$0.4464F = ?$

$= \frac{0.4464\ F \times 64g}{2\ F}$

$= 14.28\ g$ **(02 Mark)**

<u>Mass of Copper = 14.28g</u>

5.(a) i) The loss of mass was due to loss of CO2 to the surroundings. **(1marks)**

ii) $CaCO_3\ (S) + 2HCl\ (aq) \rightarrow CaCl_{2\ (aq)} + H_2O_{\ (l)} + CO_{2\ (g)}$ **(2 marks)**

iii) Mass remained constant because the reaction ended since the marble was finished. **(1 marks)**

(b) i) Water cycle

(ii) Steam engines

(iii) Refrigeration / Air condition

(iv) Refinery

(v) Drying of materials

(vi) Metallurgy **any 5 points (1 @ marks)**

6. (i) Ammonia gas**(2 marks)**

(ii) $NH_4Cl\ (s) + Ca(OH)_2\ (s) \xrightarrow{heat} CaCl_2(aq) + 2H_2O\ (l) + 2NH_3(g)$**(2 marks)**

(iii) Because is soluble in water**(1 marks)**

(iv) When a glass rod is dipped in concentrated Hydrochloric acid is placed at the mouth of a gas jar Containing Ammonia dense white fumes of Ammonium chloride are formed **(2 marks)**

$NH_3(g) + HCl\ (g) \longrightarrow NH_4Cl\ (s)$ (White fumes)

(v) When ammonia gas is passed over heated copper (II) oxide the gas is oxidized to Nitrogen while the copper (II) Oxide is reduced to copper.

$3CuO\ (s) + 2NH_3\ (g) \longrightarrow 3Cu(s) + 3H_2O\ (l) + N_2(g)$**(2 marks)**

7. (a) i) The brand of Uhai is more hard**(1 mark)**

Because; it contains high amount of Ca^{2+}ions, Mg^{2+} ions and SO_4^{2-} ions**(1 mark)**

ii) Calcium makes the teeth and bones hard**(1 mark)**

iii) Chlorine**(I mark)**

(b)

Symbol	Mn	O
% Composition by mass	49.5	50.5
Its relative atomic mass	55	16
Dividing % composition by its R.A.M	$\frac{49.5}{55}$=0.9	$\frac{50.5}{16}$= 3.156
Dividing by smallest value	$\frac{0.9}{0.9}$= 1	$\frac{3.156}{0.9}$=3.5
Converting to whole cumber	1x2=2	3.5x2=7

The empirical ratio of Mn: O is 2:7

Therefore, the empirical formula is Mn_2O_7**(5 marks)**

8. (a) (i) Because when sodium is exposed in air react vigorously with water and air.**(I mark)**

(ii) Because is slightly soluble in water / less denser than air.**(I mark)**

(b) (i) solution

Data given

Mass of $FeCl_3$ = 16.25g

Molar mass of $FeCl_3$

Given Fe = 56 Cl = 35.5

Molar mass of $FeCl_3$ = (56 + 35.5 x 3)

= (56 + 106.5) g/mol

= 162.5 g/mol

Number of moles$FeCl_3 \frac{Mass}{Molar\ mass} = \frac{16.2g}{162.5g/mol}$

= 0.1 molof $FeCl_3$

Then $FeCl_3 \longrightarrow Fe^{3+} + 3Cl^-$

Number of ions = (1 + 3) = 4

1 mole of $FeCl_3$ = 4 mol of Ions

0.1 mol of $FeCl_3$ = ?

$= \frac{4 moles\ of\ Ion \times 0.1\ mol}{1\ mol}$

= 0.4 mol of ions

$L_A = 6.02 \times 10^{23}$

Number of particles = n x L_A

= 0.4 mol x 6.02×10^{23}

= 2.408×10^{23} ions **(3.5 mark)**

Number of ions present = 2.408×10^{23} ions

(ii) Solution

Mass of H_2O = 1.8g

Molar mass of H_2O = ?

Given H = 1 O = 16

Molar mass of H_2O = 1 x 2 + 16

= 2 + 16

= 18g/mol

Number of moles$H_2O = \frac{Mass}{Molar\ mass} = \frac{1.8g}{18g/mol}$

= 0.1 mol

$L_A = 6.02 \times 10^{23}$

Number of particles (molecules) = n x L_A

= 0.1 mol x 6.02×10^{23}molecules

= 0.602×10^{23} molecules **(3.5 mark)**

Number of molecules present = 6.02×10^{22} molecules

9. (a) The type of pollution shown in the figure is water pollution. **(2 marks)**

(b) Causes of this type of pollution.

i. Pesticides
ii. Fertilizer
iii. Oil spills
iv. Mining
v. Sediments
vi. Industrial processes
vii. Sewage
viii. Poisonous gases and metals
ix. Marine debris

Any four (4) points **(1@= 4marks)**

(c) The effects of this type of pollution.

i. Spread of water born disease
ii. Eutrophication due to nutrient pollution
iii. Death of underwater organisms
iv. Loss of quality of water.
v. Exposure to oil or its constituent chemicals can alter the ecology of aquatic habitats and physiology of marine organisms
vi. Mercury has serious effects on human health, especially young children and unborn babies

Any three (3) points**(1.5 @ = 4.5 marks)**

(d) Measures that should be taken to prevent this type of pollution

i. Reducing the use of fertilizers and pesticides
ii. Treating sewage and industrial waste

iii. Stopping deforestation
iv. Controlling coastal development
v. Reducing pollution from oil spills
vi. Reducing the use of mercury

vii. Cleaning up existing and abandoned mines
viii. Cleaning up pollutants.

Any three (3) points **(1.5 @ = 4.5 marks)**

10. (a) stages in the extraction of less reactive metals includes:**(1.5@= 6 marks)**
i. Mining and concentration of ore
ii. Roasting the ore in air
iii. Reduction
iv. Purification

(b) The ores that are commonly used in the extraction of iron metals include: **(0.5@= 02marks)**
i. Haematite (Fe_2O_3)
ii. Magnetite (Fe_3O_4)
iii. Spatic iron (FeS)
iv. Iron pyrite ($FeCO_3$)

(c) i) Two reducing agents in the blast furnace include;
-Carbon monoxide (CO)
- Carbon (coke)**(01@= 02 marks)**

ii. Step 1: $C + O_2 \longrightarrow CO_2$
It is of great importance because it leads to production of carbon dioxide which later combines with carbon to form carbon monoxide, which the acts as a reducing agent.
Step 2: $CO_2 + C \longrightarrow 2CO$
It is important since it leads to production of CO (carbon monoxide) which later reduces Fe_20_3 to form Fe.
Step 3: $Fe2O3 + 3C \longrightarrow 2Fe + CO$
It is important since carbon acts as a reducing agent where it reacts with Fe_2O_3 to form iron**(03 marks)**

iii. Substance L is Limestone ($CaCO_3$)**(1 mark)**
iv. $CaCO_3CaO + CO_2CaO + SiO_2CaSiO_3$ (Slag)**(1 mark)**

11. Introduction
- ✓ Good introduction by giving out the meaning of plant nutrients, the nutrients that are required in large quantity**(2 marks)**
- ✓ Maize plant may lose its nutrients from the soil in any of the following ways;
  i. Soil erosion
  ii. Monocropping
  iii. Overgrazing and overstocking
  iv. Deforestation
  v. Burning of vegetation
  vi. Leaching
  vii. Poor farming methods
  Six point with good explanations**(12 @ 2 marks)**
- ✓ Any good conclusion **(1 mark)**